PRIVACY, DATA PROTECTION & SURVEILLANCE

XV. INTERDISCIPLINARY WORKSHOP
18 JULY 2023

PROGRAM

12:00 **Welcome and introduction round**

12:30 *Nicholas Martin (Fraunhofer ISI, Karlsruhe)*
The Chimera of Control: Some Critical Reflections on SSI's Promise to Give Users Control over their Data and Identity

13:30 *Ziyin Li (Zhejiang University; University of Copenhagen)*
Self-made Surveillance: The SWM Livestreaming and Digital Monitored Performance

14:30 **Coffee break**

14:45 *Corbinian Ruckerbauer (Stiftung Neue Verantwortung, Berlin)*
Algorithmische Entscheidungssysteme und ihre Auswirkungen auf Good Governance im Sicherheitssektor

15:45 *Wilson Guilherme Dias Pereira, Victor Barbieri Rodrigues Vieira & Paulo Rená da Silva Santarém (Instituto de Referência em Internet e Sociedade (IRIS), Belo Horizonte)*
Encrypted communication investigation: 3 unneeded methods

16:45 *Vasilis Ververis (Weizenbaum Institute, Berlin)*
An overview of Internet censorship in the EU

17:45 **Conclusion and outlook**

18:00 **End**

Location:

Alexander von Humboldt Institute for Internet and Society
Französische Straße 9
10117 Berlin
Room »Kosmos«
https://www.openstreetmap.org/#map=19/52.51452/13.38679

Online:

https://us06web.zoom.us/j/84763500539?pwd=T0I2c0hFbTRTUFRVR0Vud0RYdGFyQT09

The Chimera of Control: Some Critical Reflections on SSI's Promise to Give Users Control over their Data and Identity

Nicholas Martin (Fraunhofer ISI, Karlsruhe)

In recent years, self-sovereign identity solutions (SSI) have attracted a lot of attention. Even a cursory analysis suggests that during the Horizon 2020-funding period, European Union funding for SSI-related projects ran to ~€45 million. Member states too are funding SSI-related projects with substantial sums, and numerous startups, in Europe and abroad, are developing SSI-style identity solutions. Major business groups too have expressed strong support for SSI, especially in the financial sector (Deutscher Sparkassen- und Giroverband et al. 2021). International Organisations like the World Food Program have also explored the potential of SSI for working with refugees.

Advocates of SSI claim that it offers several benefits to users and service providers compared to traditional centralised or federated identity solutions. These are said to include greater security and usability (Kubach et al. 2020), and stimulating the digital economy (Echikson 2020). At the more optimistic end of the spectrum, proponents of SSI have even claimed that the technology could help solve the problem lack of identity documentation poses for undocumented migrants and/or support economic inclusion (Lim 2020, Wang and De Filippi 2020). The central benefit of SSI that its proponents reference is, however, that this technology would put users "in control" of their data and identity, and that this greater control over one's own data and identity is what fundamentally distinguishes SSI from other identity solutions. (Hence also the name "self-sovereign identity").

This paper critically analyses the claim that SSI gives users meaningfully greater control over (i) their data, and (ii) their identity. It argues that while not without grains of truth, these claims are, at best, greatly overblown. SSI can arguably offer users small increases in control over their data and identity, but SSI solutions do not and cannot eliminate the larger imbalances of power in the (digital) economy and the state, that are at the heart of users' lack of control over their data and identity. SSI proposes a technological and architectural solution to what are ultimately political-economic and institutional problems, and accordingly it falls short. For this reason, it also cannot solve the political problem facing most undocumented migrants, that no state is willing to issue them with valuable identity rights – contrary to some of its advocates' wilder claims.

Overblown claims are nothing new in the tech world. However, the neglect in the SSI discourse of political-economic power and the importance of institutions, rather than technology, for ameliorating power imbalances and protecting users, leads to a limited awareness of how risks to users in fact tend to arise: namely from the interplay of technology, institutions and power. The same technology or architecture that in one political-institutional setting can increase user control, may diminish it in another. Specifically,

absent institutional safeguards, power imbalances between users and service providers can transform the architecture of SSI from a (mildly) privacy-enhancing technology into a "disclosure machine", where users are forced to give up more data about themselves than they would likely have to (or be able to) with either centralised or federated identities. This problem emerges particularly clearly with refugees.

Beyond the potential of SSI schemes to increase, rather than decrease, users' transparency and data disclosure, the somewhat naïve rhetoric about giving users' "control" over their data and identity also endangers SSI adoption itself. As surveys attest, among users, the promise of getting greater control over their data is very popular (PWC 2021, Spiekermann and Korunovska 2017). It is likely for this reason that business interests, who may have more prosaic and self-interested economic reasons to promote SSI, have adopted the language of giving users "control over their data" to advocate for SSI (Deutscher Sparkassen- und Giroverband et al. 2021; Schorlemer 2022). However, if this paper's contention, that the "control" SSI grants to users is largely chimerical, is correct, it is to be expected that users will soon enough see through this promotional rhetoric. That would be bad for SSI – which arguably does offer users real economic and usability benefits – and for the general trust in society.

The paper proceeds as follows. Section 2 analyses the concept of having "control" over one's data and identity. Section 3 describes how SSI solutions try to provide users with "control" by technological and architectural means. Section 4 analyses why this promise of SSI largely falls short, with the (distinctly modest) improvements in user "control" that SSI solutions may sometimes be able to achieve depending entirely on the existence of supportive legal structures. Section 5 then analyses the risk SSI presents when such legal structures are lacking, namely to turn into a "disclosure machine" – the very opposite it is meant to achieve. Section 6 concludes.

Self-made Surveillance: The SWM Livestreaming and Digital Monitored Performance

Ziyin Li (Zhejiang University; University of Copenhagen)

> *When I was in high school, we studied for exams. There was a CCTV camera in the classroom and I sat next to the podium, so I was uncovered. If I weren't focused on studying, but reading a novel, eating snacks, or dozing off, the teacher would find out and be mad. I think the live streaming has helped me find a similar atmosphere as I had that time. (1)*

Live streaming is a modern practice for Generation Z. Not only watching and interacting, they started using it for self-tracking and self-regulating their lives. "StudyWithMe" live streaming (SWM), which emerged from YouTube, has seen a boom in live streaming sites in China. Reportedly, among all live streaming genres on Bilibili (a comprehensive live streaming platform), SWM has the longest duration of single streaming, usually over 6 hours. During 2018, over 1.03 million SWM were started on Bilibili, with a total duration of 1.46 million hours and an average of nearly 2,000-3,000 daily sessions. Popular SWM has hundreds of thousands of subscribers.

How is it different from other self-tracking tools? The self-explanatory fact is SWM has online cameras. While the cameras are always seen as a tool by surveillance theories, sometimes with the productivity of course, first, why the streamers as students or examinees chose SWM to self-regulate? Second, how does SWM affect their learning? And in general, does the specific practice in certain contexts echo the self-surveillance theory, which emphasizes the recording and subsequent reflection of an individual's own behavior, or bring new perspectives to it?

We did online observation in 48 randomly chosen SWM rooms from Bilibili from October 2021 to May 2022 for gathering the brief overview. And We use semi-structured interviews as the main research method. 15 Bilibili SWM streamers have been interviewed between December 2021 to February 2022.

Firstly, they use SWM streaming as an efficient tool to regulate themselves since it can recreate a context of learning online. The streamers see themselves as the main characters and model for the imaginary audience. Meaning that they are afraid to play, chill or not study in front of the camera, or even leave the camera view for too long. Some also tried to expand the area that the camera can shoot to prevent themselves from being lazy. Comments and interactions from actual audiences also provide real regulation. For example, the audience sometimes checks if the streamer is studying as the timetable (on-screen page) planned.

One cannot argue if the streamers feel SWM is a productive tool for self-learning. But things started to be extreme, from what they say. The "Student Regulations" of Chinese education constitute, first and foremost, the streamer's self-examination rules. They must do things as required, only study but not dance or take gifts even though it's quite com-

mon in other types of streamings. They need to ask for leave and make up the originally planned live (study) hours. They even need to face the "ultimate scrutiny", meaning they have to show the results of the exam they prepared for, which will almost always determine the final evaluation of the streamer. Some also try to extend their study hours as much as possible to be perfect SWM learners.

In China, social roles for students or streamers are both pretty clear, and they are combined now into SWM streaming. Slightly compared with the SWM on YouTube, Chinese SWM can last 6 to 8 hours, or even all day and night. It blurs the line between digital performance and real life. Streamers actively live in this cultural practice, based on the logic of the Chinese exam-oriented education model. Meanwhile, they also take the role of a streamer (or an influencer) and want to gain good streaming views, likes, and comments, even making money from it.

When Humphreys (2011) proposed "self-surveillance as an extension of self-monitoring", it implied that the normative forces imposed on the individual by the specific cultural context. And in this case, the fact of being monitored changes the meaning of the performance. In the continuous spontaneous viewing, retrieval and analysis, the monitored performance constantly confirms and reinforces the power of a certain, or combined, cultural norm, causing a continuous compression on the individual. The streamers not only have to do the action itself but also need to cater to the audience's orientation, which strongly depends on the local culture, by launching into a physical and verbal performance. It means that, learning in streaming, the individual need to become a "perfect" student; Working in streaming, the individual becomes a "perfect" laborer. As the streaming become longer and longer, and each day becomes a live day, the deep backstage beyond the role stack is not only compressing but perhaps will eventually cease to exist, replaced by progressively deeper surveillance and role discipline. And today it is embraced, led, and even initiated by the individuals themselves and spills over into the digital space.

Encrypted communication investigation: 3 unneeded methods

Wilson Guilherme Dias Pereira, Victor Barbieri Rodrigues Vieira & Paulo Rená da Silva Santarém (Instituto de Referência em Internet e Sociedade (IRIS), Belo Horizonte)

This workshop session will delve into three so-called "alternative" methods for investigating private encrypted communications: instant messages traceability, client-side scanning, and government hacking. The objective is to provide information on the approaches employed by law enforcement agencies worldwide, which supposedly aim to preserve encryption, protect human rights, and enhance investigation efficiency. We will specifically focus on the Brazilian perspective regarding these global issues. During the session, we will present a panoramic synthesis of three reports published in 2022, enabling participants to gain valuable insights into the investigation of private encrypted communications and the accompanying legal, technical, and ethical considerations. By the end of the workshop, attendees will have a well-rounded understanding of these three methods and their implications.

An overview of Internet censorship in the EU

Vasilis Ververis (Weizenbaum Institute, Berlin)

By establishing an infrastructure for monitoring and blocking networks in accordance with EU law on preventive measures against the spread of information, EU member states have also made it easier to block websites and services and monitor information. While relevant studies have documented Internet censorship in non-European countries, as well as the use of such infrastructures for political reasons, this study examines network interference practices such as website blocking against the backdrop of an almost complete lack of EU-related research in the EU.

Specifically, it performs and demonstrates an analysis for the total of 27 EU countries based on three different sources. They include first, tens of millions of historical network measurements collected in 2020 by OONI volunteers from around the world; second, the publicly available blocking lists used by EU member states; and third, the reports issued by network regulators in each country from May 2020 to April 2021.

Our results show that authorities issue multiple types of blocklists. ISPs limit access to different types and categories of websites and services. Such resources are sometimes blocked for unknown reasons and not included in any of the publicly available blocklists. The study concludes with the hurdles related to network measurements and the non transparency from regulators regarding specifying website addresses in blocking activities.

Corbinian Ruckerbauer

Algorithmische Entscheidungssysteme und ihre Auswirkungen auf Good Governance im Sicherheitssektor

Extended Abstract für den XV. Interdisziplinären Workshop Privacy, Datenschutz & Surveillance am 18.07.2023

Sicherheitsproduktion in der datafizierten Gesellschaft

Der Einsatz digitaler Technologien in allen Lebensbereichen führt zu einer rasant wachsenden Menge an Daten. Nahezu jedem Subjekt, jedem Objekt und jeder Handlung, die in ihrer Summe unsere Gesellschaften konstituieren, wird ein Datum zugeordnet. Dies hat zur Folge, dass die bei Verfügbarkeit geeigneter Technologien und Daten potenziell generierbaren Erkenntnisse beinahe unbegrenzt sind. Diese Entwicklung berührt dabei den Kern des Aufgabenbereichs von Sicherheitsbehörden zur Gefahrenabwehr und -prävention. Stetig an Bedeutung gewinnt die technologische Fähigkeit, aus einer großen, für Menschen nicht erschließbare, Menge von Daten relevante Informationen zu extrahieren. Immer leistungsfähigere algorithmische Systeme werden außerdem eingesetzt, um auf Grundlage dieser Datenmengen optimierte Entscheidungen zu ermitteln.

Dem Vortrag liegt dabei die Frage zugrunde, wie sich die Verwendung solcher algorithmischer Entscheidungssysteme auf Prinzipien der *Good Governance* im Sicherheitssektor auswirkt. Um diese Frage zu beantworten, sollen zunächst zentrale Aspekte von *Good Governance* im Sicherheitssektor diskutiert werden: die Gewährleistung der Rechenschaftspflicht der Akteure, Rechtstaatlichkeit, Transparenz, Effektivität und Effizienz. Anschließend soll ein Überblick über die Anwendungsfelder algorithmischer Entscheidungssysteme in unterschiedlichen Teilbereichen des Sicherheitssektors gegeben werden. Ausgehend von diesen Anwendungsfeldern werden dann Auswirkungen auf die zuvor besprochenen Aspekte der *Good Governance* im Sicherheitssektor diskutiert,

bevor Handlungsleitlinien aus der aktuellen Gesetzgebung und Rechtsprechung für den Einsatz diskutiert werden.

Unterschiede bei der Verwendung algorithmischer Entscheidungssysteme bestehen hinsichtlich der Funktion des jeweiligen Systems im Entscheidungsprozess. Sie können von Menschen im Entscheidungsprozess als einer der relevanten Faktoren berücksichtigt werden, den menschlichen Entscheidungsspielraum einschränken oder ohne weiteren menschlichen Einfluss zu einer Entscheidung führen.

Systemübergreifende Datenanalysetools verwenden Algorithmen, um die stetig wachsenden und oft fragmentierten Datenmengen beispielsweise von Strafverfolgungsbehörden zusammenzuführen, diese zu analysieren und so Informationen zu extrahieren, die ohne diese Technologie unmöglich oder nur mit hohem Ressourcenaufwand zu ermitteln wären.

Wirkung auf Good Governance im Sicherheitssektor

1) Effizienz & Effektivität

Durch die Möglichkeit in Bruchteilen von Sekunden große Mengen von Daten zu verarbeiten, die vorher enorme Personalressourcen in Anspruch genommen hätten und die Fähigkeit, Informationen und Zusammenhänge zu konstruieren, die Menschen nicht in der Lage wären zu erkennen, erhöhen sich Effizienz und Effektivität in der Sicherheitsproduktion.

Diese positive Wirkweise hängt dabei von mehreren Faktoren ab: der Qualität der Daten, der Qualität der Systeme und der Eignung der Technologie für den konkreten Anwendungsfall. Abhängig von diesen Faktoren kann sich der Einsatz auch nachteilig auswirken, nämlich beispielsweise dann, wenn Ermittlungsansätze auf fehlerhaften oder unzureichenden Daten basieren. Eine weitere Herausforderung besteht darin, dass es sich bei algorithmischen Entscheidungssystemen häufig um proprietäre Software handelt, die von privatwirtschaftlichen Unternehmen entwickelt wird. Dies kann zu Lock-in Effekte führen. Selbst wenn Sicherheitsbehörden feststellen, dass eine andere technologische Lösung effektiver wäre, wird diese dann möglicherweise nicht umgesetzt.

2) Rechtstaatlichkeit und der Schutz von Grundrechten

Entscheidender Bestandteil der Beurteilung der Rechtmäßigkeit des Einsatzes algorithmischer Entscheidungssysteme und der damit verbundenen Erfassung und Verarbeitung personenbezogener Daten ist die Verhältnismäßigkeitsprüfung. Dabei sind

die Wirksamkeit der Maßnahme, die Intensität des Eingriffs und der verfolgte Zweck zu berücksichtigen. Die fragliche Effektivität solcher Systeme, beispielsweise im Bereich der Gefahrenabwehr, wirft somit auch Fragen hinsichtlich der Rechtmäßigkeit des Einsatzes auf.

Unabhängig von der Verhältnismäßigkeitsprüfung der Maßnahmen wirft der Einsatz algorithmischer Entscheidungssysteme Fragen hinsichtlich der Wahrung von Grundrechten auf, wie zum Beispiel die Unschuldsvermutung, das Recht auf einen fairen Prozess und das Recht auf einen wirksamen Rechtsbehelf. Eine zentrale Problematik liegt dabei auch in der reduzierten ‚Sichtbarkeit' von Grundrechtseingriffen für Betroffene und kontrollierende Akteure, wenn diese durch Datenverarbeitung vorgenommen werden.

Andererseits können algorithmische Entscheidungssysteme auch dazu dienen, den Schutz von Grundrechten zu stärken. Durch den Einsatz geeigneter und grundrechtskonformer Systeme, die bestehende Diskriminierungen nicht reproduzieren, kann die Voreingenommenheit gegenüber bestimmten Gruppen im menschlichen Handeln als Fehlerquelle ausgeschlossen werden.

3) Rechenschaftspflicht & Transparenz

Zudem können solche digitalen Anwendungen eine Stärkung der Rechenschaftspflicht ermöglichen. So erleichtern sie die Evaluation der Arbeit von Akteuren im Sicherheitssektor, um individuelles oder systematisches Fehlverhalten sowie Diskriminierungen zu erkennen. Insbesondere die Möglichkeit, Log-Dateien aufzuzeichnen und diese mit algorithmischen Methoden auszuwerten, stärkt die Effektivität der Kontrolle durch Aufsichtsbehörden.

Diese Effektivität der Kontrolle wird jedoch gleichzeitig durch die zentrale Rolle privatwirtschaftlicher Akteure herausgefordert. Durch proprietäre, nicht quelloffene Software kann die Kontrollierbarkeit bestimmter Anwendungen und ihrer Wirkung ebenso untergraben werden, wie die Transparenz des Handelns der Sicherheitsbehörden. Diese Schwächung der Rechenschaftspflicht nimmt zu, wenn diese privatwirtschaftlichen Akteure in Drittstaaten ansässig sind. Zum einen kann dies die Gewährleistung von Grundrechten unterlaufen, wenn beispielsweise die Daten in den jeweiligen Staaten nicht gleichermaßen vor dem Zugriff der dortigen Sicherheitsbehörden geschützt sind, zum anderen erschwert dies die Kontrolle durch die jeweiligen Behörden.

Zusammenfassend ist festzustellen, dass der Einsatz algorithmischer Entscheidungssysteme im Sicherheitssektor erhebliche Auswirkungen auf Prinzipien der

Good Governance hat. Es ist entscheidend, geeignete Handlungsleitlinien zu entwickeln, um die positiven Effekte zu nutzen und gleichzeitig Risiken zu reduzieren. Ausgehend von den ermittelten Herausforderungen bei der Verwendung von algorithmischen Entscheidungssystemen sollen abschließend einige Handlungsleitlinien für den Umgang mit diesen Systemen im Sicherheitssektor erläutert werden.